DAY 23 – AUTHENTICATION LOG ANALYSIS

Cyber Security Internship – Veda Technology

1. Objective

To analyze synthetic authentication logs using Python and Excel, identify successful and failed login attempts, and examine authentication patterns based on usernames and source IP addresses. The task focuses on developing basic security log analysis skills and identifying potentially suspicious login activities.

2. Introduction

Authentication is the process of verifying the identity of a user before granting access to a system, application, or network.

Authentication logs record login activities, including successful logins, failed attempts, usernames, timestamps, and source IP addresses.

Analyzing these records helps cybersecurity professionals identify unusual login behavior, repeated authentication failures, and possible unauthorized access attempts.

Python provides an efficient way to process authentication logs, while Excel helps organize and visualize the results.

Common Authentication Log Fields

Field	Description
Timestamp	Date and time of the login attempt
Username	Account used for authentication
Source IP	IP address associated with the login attempt
Event Type	Authentication activity performed
Status	Indicates success or failure
Login Method	Method used for authentication, such as SSH

3. Tools and Technologies

Tool	Purpose
Python	Processing and analyzing authentication logs
Pandas	Filtering, grouping, and summarizing log data
Matplotlib	Visualizing authentication patterns
Excel	Reviewing log records and analysis tables
CSV Dataset	Storing synthetic authentication events

4. Dataset Description

A synthetic authentication dataset containing 12 login events is used to demonstrate authentication log analysis.

The dataset includes different usernames, source IP addresses, timestamps, and login statuses.

Sample Authentication Dataset

Time	Username	Source IP	Status
09:00	admin	192.0.2.10	Failed
09:01	admin	192.0.2.10	Failed
09:02	admin	192.0.2.10	Failed
09:03	admin	192.0.2.10	Failed
09:05	yuktha	192.0.2.20	Success
09:10	intern1	192.0.2.30	Success
09:12	root	198.51.100.15	Failed
09:13	root	198.51.100.15	Failed
09:15	guest	203.0.113.25	Failed
09:20	yuktha	192.0.2.20	Success
09:25	intern1	192.0.2.30	Success
09:30	admin	192.0.2.10	Failed

All events are dated October 23, 2026. The IP addresses belong to documentation ranges and are used only for simulation.

5. Practical Implementation

Step 1: Import Required Libraries

```
import pandas as pd

import matplotlib.pyplot as plt
```

Description:

Pandas is used to organize and analyze authentication records, while Matplotlib is used to create visual representations of login activities.

Step 2: Load the Authentication Dataset

```
df = pd.read_csv("authentication_logs.csv")

print(df.head())
```

Description:

The CSV file is loaded into a Pandas DataFrame. The head() function displays the first five authentication records.

Step 3: Count Successful and Failed Logins

```
login_counts = df["Status"].value_counts()

print(login_counts)
```

Expected Result:

Login Status	Count
Failed	8
Success	4
Total	12

Observation:

The synthetic dataset contains eight failed login attempts and four successful logins.

A higher number of failures may indicate incorrect credentials or potentially suspicious authentication behavior.

Step 4: Analyze Login Attempts by Username

```
user_analysis = pd.crosstab(
    df["Username"],
    df["Status"]
)

print(user_analysis)
```

Expected Result:

Username	Failed	Success	Total
admin	5	0	5
guest	1	0	1
intern1	0	2	2
root	2	0	2
yuktha	0	2	2
Total	**8**	**4**	**12**

Observation:

The admin account recorded the highest number of failed login attempts, with five failures.

Repeated failures against administrative accounts require additional investigation.

Step 5: Analyze Authentication Events by Source IP

```
ip_analysis = pd.crosstab(
    df["Source IP"],
    df["Status"]
)
print(ip_analysis)
```

Expected Result:

Source IP	Failed	Success
192.0.2.10	5	0
192.0.2.20	0	2
192.0.2.30	0	2
198.51.100.15	2	0
203.0.113.25	1	0

Observation:

The source IP 192.0.2.10 generated five failed login attempts targeting the admin account.

This pattern could indicate password-guessing activity, although further evidence would be required to confirm an attack.

Step 6: Identify Repeated Failed Login Attempts

```
failed = df[df["Status"] == "Failed"]


suspicious = (
    failed.groupby(["Username", "Source IP"])
    .size()
    .reset_index(name="Failed Attempts")
)
```

```
suspicious = suspicious[
    suspicious["Failed Attempts"] >= 3
]


print(suspicious)
```

Expected Result:

Username	Source IP	Failed Attempts
admin	192.0.2.10	5

Description:

The program groups failed authentication attempts by username and source IP address.

A demonstration threshold of three or more failures is used to identify patterns requiring investigation.

Step 7: Visualize Authentication Results

```
login_counts.plot(
    kind="bar",
    title="Authentication Login Analysis"
)


plt.xlabel("Login Status")
plt.ylabel("Number of Attempts")
plt.tight_layout()
plt.show()
```

Description:

A bar chart is generated to compare successful and failed login attempts.

The visualization makes it easier to identify the distribution of authentication outcomes.

6. Authentication Pattern Analysis

The synthetic dataset was examined to identify unusual login activities.

Pattern	Observation	Security Concern
Repeated failed logins	Five failures for admin	Possible password guessing
Root login failures	Two failed attempts	Attempted access to privileged account
Successful logins	Four successful events	Require authorization verification
Multiple source IPs	Five unique IP addresses	Different authentication sources
Consecutive failures	Four failures within three minutes	Potential automated login attempts

Suspicious Activity Indicators

1. Multiple failed login attempts targeting the same username.
2. Repeated failures originating from a single source IP.
3. Authentication attempts targeting privileged accounts.
4. Unusual login frequencies or unexpected access times.
5. Successful authentication following repeated failures, which may warrant closer examination.

These indicators do not independently confirm malicious activity but help prioritize security investigations.

7. Statistical Summary

Parameter	Result
Total Authentication Events	12
Successful Logins	4
Failed Logins	8
Unique Usernames	5
Unique Source IPs	5
Failed Login Percentage	66.67%
Successful Login Percentage	33.33%
Highest Failed Attempts	5

The dataset shows that failed authentication attempts account for approximately 66.67% of all recorded events.

This highlights the importance of monitoring authentication failures and investigating repeated unsuccessful login patterns.

8. Security Recommendations

Identified Concern	**Recommended Security Measure**
Repeated failed logins	Configure login rate limiting and security alerts
Administrative account targeting	Restrict privileged account access
Weak authentication	Implement strong passwords and MFA
Suspicious source IP activity	Investigate source IPs and apply appropriate network controls
Unauthorized access attempts	Monitor authentication logs regularly
Lack of monitoring	Implement centralized log analysis

9. Importance of Authentication Log Analysis

Authentication log analysis provides several cybersecurity benefits:

1. **Threat Detection:** Helps identify suspicious authentication attempts.
2. **Account Protection:** Detects repeated failures against sensitive accounts.
3. **Incident Investigation:** Supports the reconstruction of authentication-related events.
4. **Access Monitoring:** Helps verify whether user login activity is authorized.
5. **Security Auditing:** Provides records for reviewing authentication controls.
6. **Early Warning:** Helps identify potential attacks before they escalate.

10. Results and Observations

The illustrative Python analysis demonstrates how authentication logs can be grouped, filtered, and examined to identify suspicious login patterns.

Analysis Activity	**Result**
Dataset Preparation	12 synthetic authentication records
Login Status Analysis	8 failed and 4 successful
Username Analysis	admin had the most failed attempts
Source IP Analysis	192.0.2.10 had five failures

Suspicious Activity Detection	One username-IP combination met the threshold
Visualization	Bar chart comparing login outcomes
Security Assessment	Potential password-guessing pattern identified

The results demonstrate how basic data analysis techniques can support cybersecurity monitoring and help identify authentication events requiring further investigation.

11. Conclusion

The Authentication Log Analysis task demonstrated the importance of monitoring user login activities to identify possible security threats.

Using Python and Pandas, synthetic authentication records were analyzed to determine successful and failed login counts, examine user-specific activities, and group authentication attempts by source IP addresses.

The analysis identified repeated failed login attempts targeting the admin account, demonstrating how suspicious authentication patterns can be detected through log analysis.

Overall, the task strengthened foundational skills in Python-based security data analysis, authentication monitoring, and threat identification, which are important for cybersecurity and Security Operations Center (SOC) activities.